Individual Assignment Activity: Requirements Analysis
Rizalina A. Ortega
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3088 - Information System

Module assigned : Treasurer's Office Tax Collection and Billing Management Module

1. Project overview and stakeholder list

- **Overview**
  - Our project group is developing an automated web platform titled as ***Needs Assessment for a Real Property Assessment Tracking and Inter-Office Synchronization System: Resolving Disconnected Records and Workflow Delays in Municipal Assessor and Treasurer Offices in Pasacao.*** The main goal of our study is to conduct a needs assessment and comparative evaluation of local municipal workflows across Canaman, Magarao, San Fernando and benchmarking with Pasacao (which uses an existing eRPT setup) to solve real-world problems in local government units. Specifically, our project addresses the severe administrative frictions, valuation delays, fragmented record-keeping, and the massive operational headache of migrating decades-old legacy records dating back to 1949.

    Building on this overall group project, my assigned module is the **Treasurer's Office Tax Collection and Billing Management Module**. This module receives the Assessment Roll from Module 1, generates Real Property Tax billing statements, records payments, tracks delinquent accounts, and issues tax clearances. The module addresses three gaps found during field interviews: the Treasurer's Office receives Assessment Roll data only through manual submission from the Assessor's Office, no digital billing statement is issued to taxpayers before the collection window, and there is no automated system to flag and track delinquent properties.

- **Stakeholders and Interest / Concerns**
  - **Municipal Treasurer (Canaman, Pasacao)**
    - Roles and Interest : Manages real property tax collection. Needs an accurate billing ledger, automated computation of discounts and penalties, and a delinquency tracking list.
  - **Treasurer's Office Staff and Collection Clerks**
    - Roles and Interest: Process walk-in payments and issue tax clearances daily. Need fast taxpayer record lookup, pre-computed tax bills, and payment receipt generation.
  - **Real Property Owners and Taxpayers**

        - Roles and Interest: Pay real property tax annually or in quarterly installments. Need a clear billing statement showing the amount due, applicable discount or penalty, and payment options.
    - **IT Developer and Capstone Team**
        - Designs, builds, and evaluates the system against ISO/IEC 25010.

2. Information-gathering methods used and summary of findings

To gather these insights, our group used several information-gathering methods, including field interviews with municipal personnel in Canaman and Pasacao, analysis of official documents and assessment records, comparative observations between manual and eRPT workflows, and consultations with our research advisers.

Methods Used

- **Field Interviews:** The team conducted structured interviews with the Assessor's Office staff at Canaman and Pasacao. Those interviews revealed details about the workflow connection between the Assessor's Office and the Treasurer's Office. A separate interview session at the Canaman office included a Treasurer's Office staff member who described the collection process.
- **Document Analysis:** The team reviewed the discount and penalty schedule as described by interviewees. The team also reviewed the Assessment Roll submission process as described by the Canaman Assessor.
- **Comparative Observation** : The team compared the manual collection workflow of Canaman (paper ledgers and walk-in counter payments) against the described workflow of Pasacao, where the Assessor's Office uses an eRPT system that computes the Assessed Value used as the basis for tax billing.
- **Professor Consultations:** We consulted our project to the professors for guidance on our research direction. Their main advice was that our project must focus first on fixing local office connections linking the local assessor and treasurer offices so they share information smoothly before trying to solve complex provincial Capitol approval delays.

Key Findings

From Canaman (Manual Baseline)
• **Office Coordination:** The Canaman Assessor confirmed that the Assessor's Office provides property records and Assessment Roll data to the Treasurer's Office, noting that tax collection is handled entirely by the Treasurer. • **Discounts and Penalties:** The office follows the discount and penalty structure under Republic Act No. 7160: a 20% discount for advance payment by December of the current tax year, a 10% discount for payment within the first quarter, and a 2% per month penalty for late payments.

- **Back-Year Collections:** Taxpayers frequently arrive at the counter with very old records to pay uncollected taxes, requiring the Treasurer's Office to manually handle back-year billing and accumulated penalty computations.
- **Treasury Billing & Payment Process:** A Treasurer's Office staff member confirmed that the office computes tax bills, collects counter payments, and issues receipts.
- **Irregular Payment Patterns:** Staff noted that many taxpayers only pay when they need a tax clearance for transactions like property transfers or financial assistance, making delinquency tracking difficult without a centralized system.
- **Assessment Basis & Installments:** The treasury uses the Assessed Value from the Assessor's Assessment Roll for manual tax computations. Furthermore, payments can be made in four equal quarterly installments as prescribed by Republic Act No. 7160, Section 250 (due by March 31, June 30, September 30, and December 31).

From Pasacao (Partially Digitized Benchmark)

- **System Integration:** The Pasacao tax mapper confirmed that their eRPT system automatically computes the Assessed Value and passes it to the Treasurer's Office via the Assessment Roll. Both offices share data so the treasurer can track tax payments.
- **Lack of Online Portals:** Similar to manual offices, taxpayers in Pasacao often wait until they need a clearance to pay, and there is no digital billing notification or status portal available for public use, requiring physical visits to check balances.

From Professor and Adviser Consultations

- **Scope Confirmation:** The Capstone Professor confirmed that tax collection and payment processing belong strictly to the Treasurer's Office scope. He validated that automated billing, statement generation, and delinquency tracking are valid primary functions for Module 2.
- **Payment Gateways:** He noted that online payment via API integration is a future enhancement and is not required for this academic year.
- **Data Flow:** The professor confirmed that the Assessment Roll serves as the direct input data source for the billing computation module.
- **Systemic Gaps:** The Research Adviser confirmed that the workflow disconnection between the Assessor's and Treasurer's offices is a systemic problem across all visited municipalities, directing the team to treat this data gap as a primary deliverable.

3. Functional requirements (prioritized)

MUST HAVE

- This module must receive and import the Assessment Roll from Module 1. The import must capture, for each property: the Assessment Roll Number, Property Index

Number, Owner Name, Property Address, Property Classification, Assessed Value, and Tax Declaration effectivity date. This module must not require manual re-entry of data that Module 1 already holds.

- This module must compute the annual Real Property Tax due for each property automatically. The computation must follow Republic Act No. 7160, Section 233: Basic Real Property Tax equals Assessed Value multiplied by the applicable tax rate. For provinces, the tax rate must not exceed 1%. The computation must also include the Special Education Fund levy of 1% of the Assessed Value.
- This module must compute discounts and penalties automatically. The computation must follow Republic Act No. 7160: a 20% discount applies when the taxpayer pays by December 31 of the current tax year (Section 251); a 10% discount applies when the taxpayer pays within the first quarter of the following year (Section 251); a 2% per month penalty applies for each month of delay past the first-quarter deadline, up to a maximum of 72% (36 months), as prescribed by Section 255.
- This module must support quarterly installment payment scheduling. The four installment due dates must follow Republic Act No. 7160, Section 250: on or before March 31, June 30, September 30, and December 31 of the tax year.
- This module must generate a billing statement for each property. The billing statement must show: Owner Name, Property Address, Property Classification, Assessed Value, Basic Real Property Tax due, Special Education Fund levy, applicable discount or penalty, total amount due, and the installment breakdown.
- This module must record each payment transaction. Each record must capture: the date of payment, the amount paid, the tax year covered, the installment quarter covered, the payment type (full or installment), and the name of the collecting staff member.
- This module must issue a payment receipt. The receipt must show: the amount paid, the date, the property covered, the tax year, and the receipt number. The receipt must be printable.
- This module must allow staff to search taxpayer records by Owner Name, Property Index Number, Assessment Roll Number, and Barangay. The module must return results in five seconds or less.
- This module must generate a Tax Clearance document for a taxpayer whose account has no outstanding balance. The Tax Clearance must state the property covered, the tax year verified as paid, and the date of issuance.

SHOULD HAVE

- This module should flag all delinquent properties automatically. A property is delinquent when the first-quarter payment deadline passes without a recorded payment. The flag must show the number of months past due and the accumulated penalty amount.
- This module should maintain a complete payment history for each property. The history must show all payment transactions from the earliest recorded tax year available, including back-year payments and adjustments.

- This module should generate a delinquency list report. The report must list all properties with outstanding balances, sorted by barangay. Staff must be able to export this report in PDF or CSV format.
- This module should compute back-year tax dues automatically. When a taxpayer has unpaid taxes from prior years, the module must compute the total due including the principal for each year, the accumulated penalty at 2% per month per year, and any prior Special Education Fund levy arrears.
- This module should display an alert on the staff dashboard when the December 31 advance payment discount period is active and when the March 31 first-quarter deadline is within 30 days.

COULD HAVE

- This module could allow taxpayers to check their billing status online without visiting the office. The taxpayer must enter the Property Index Number or Assessment Roll Number to view the current amount due, payment history, and any outstanding balance.
- This module could allow barangay treasurers to record payments for properties within their barangay, following the deputization arrangement under Republic Act No. 7160, Section 247. Access must be restricted to the barangay the treasurer covers.
- This module could generate a monthly collection summary report. The report must show total collections per barangay, per property classification, and per payment type for the selected period.

WON'T HAVE

- This module will not process online payments or connect to payment gateways. The Capstone Professor confirmed this is a future API integration. It is outside the scope of this academic year.
- This module will not issue electronic fund transfer receipts or connect to any banking system. All payments are recorded as walk-in counter transactions.
- This module will not manage non-real-property taxes, fees, or other Treasurer's Office collections. This module covers real property tax only.
- This module will not replace the formal physical signing process at the Provincial Capitol in Pili. That function is the scope of Module 3.

4. Non-functional requirements (prioritized)

MUST HAVE	
Non-Functional Requirement	**Measurable Target**
Performance Efficiency: This module must return taxpayer record search results within an acceptable time. This applies to all records, including back-year payment histories.	Search response ≤ 5 seconds for any record
Functional Suitability : Tax billing computations (Basic Real Property Tax, Special Education Fund levy, discounts, penalties, and installment amounts) must match manual verification results at all times.	0% computation error rate versus manual benchmark calculations during evaluation.
Security and Data Privacy: This module must restrict access to all stored payment records and taxpayer information. Only authenticated Treasurer's Office staff may access or change payment records. This requirement complies with Republic Act No. 10173 (Data Privacy Act of 2012).	Role-Based Access Control enforced for all record access and modification.
Usability: Treasurer's Office staff with minimal technical training must complete a standard payment recording and receipt generation without external help.	ISO/IEC 25010 Usability score ≥ 4.0 out of 5.0 on the user evaluation survey.
SHOULD HAVE	
Non-Functional Requirement	**Measurable Target**
Reliability: This module must handle multiple simultaneous users without data loss or corruption. This applies during peak periods such as the first-quarter payment rush and the December advance payment window.	Module remains stable under at least five concurrent users.
Maintainability : An authorized administrator must update tax rates and Special Education Fund levy rates without changing the source code. The provincial or national authority may update these rates by ordinance or legislation.	Admin panel allows rate updates without developer intervention.
Processing Time Reduction: This module must reduce the time to complete a payment	Average payment recording and receipt generation time ≤ 10 minutes versus the

transaction and issue a receipt compared to the manual baseline.	current manual baseline of 20 to 30 minutes estimated from field interview descriptions.
COULD HAVE	
Non-Functional Requirement	**Measurable Target**
Accessibility: The taxpayer-facing billing status page must work on mobile browsers.	Responsive design functional on screen widths ≥ 360 pixels.
WON'T HAVE	
This module will not include blockchain validation of payment receipts. The Capstone Professor noted this as a possible future enhancement. It is outside the scope of this academic year.	

5. Key use cases or user stories (3–5 high-priority ones)

1. **Annual Tax Bill Generation from Assessment Roll**
    - As a Treasurer's Office staff member, I want to import the Assessment Roll from the Assessor's Office module and have the module automatically compute each property's annual tax bill, so that I can issue billing statements without manually re-entering data or computing the amounts by hand

2. **Walk-in Payment Recording and Receipt Issuance**
    - As a Treasurer's Office collection clerk, I want to record a taxpayer's payment and print a receipt immediately, so that the taxpayer leaves with official proof of payment and the record is stored in the module without a separate manual ledger entry

3. **Discount and Penalty Computation**
    - As a Treasurer's Office staff member, I want the module to automatically show the correct discount or penalty based on the current date, so that I can tell the taxpayer the accurate amount due without computing it manually at the counter.

4. **Tax Clearance Issuance**
    - As a Treasurer's Office staff member, I want to verify that a taxpayer has no outstanding real property tax balance and generate a tax clearance document instantly, so that the taxpayer can proceed with a transfer of ownership or a financial transaction without delay.

5. **Delinquency List Generation**
    - As the Municipal Treasurer, I want to generate a list of all delinquent properties sorted by barangay, so that I can identify and prioritize collection efforts for properties with unpaid real property tax

6. Assumptions, constraints, and open issues

ASSUMPTIONS
• Module 1 will produce an Assessment Roll export in a format that Module 2 can import without manual reformatting. The Assessment Roll export format must be confirmed between the two module developers before development begins. • The Municipal Treasurer and at least one collection clerk will participate in system evaluation during the pilot phase. • The team will use mock or masked taxpayer payment data during development and testing to comply with Republic Act No. 10173. The team will use actual payment data only during the supervised field evaluation phase. • Internet connectivity at the target Treasurer's Office is adequate for a web-based module. The Canaman office confirmed access to a centralized WiFi network. • Tax rates follow Republic Act No. 7160 and any applicable provincial ordinance. The team will confirm the exact tax rate in force for Camarines Sur municipalities before encoding the billing computation rules. • This module covers real property tax collection functions only. Other Treasurer's Office collections (business permits, fees, and other local taxes) are outside the scope of this module. • The Tax Clearance document generated by this module requires the Municipal Treasurer's physical signature before release during the pilot phase.
CONSTRAINTS
• **Policy:** Real property tax rates, discount percentages, and penalty rates follow Republic Act No. 7160 (Local Government Code of 1991) and applicable provincial ordinances. • **Legal:** Maximum interest penalty on delinquent real property tax must not exceed 36 months (72%) as prescribed by Section 255. • **Legal and Data Privacy:** All payment records and taxpayer information are subject to Republic Act No. 10173 (Data Privacy Act of 2012) via Role-Based Access Control. • **Time:** The academic project timeline runs from July to December 2026 for a working prototype deliverable. • **Technical:** No external payment gateway integration; all transactions are walk-in counter recordings during this phase. • **Regulatory:** Current computations follow the existing tax framework rather than the 2028 targets of Republic Act No. 12001.
OPEN ISSUES
• Will the Treasurer's Office provide an exported digital dataset of the historical tax collection records, or must the development team manually encode the outstanding

balances and payment ledgers from the physical record books into the system database?

- The system must handle payment transactions where the person at the counter is acting on behalf of the property owner. Those cases require a Special Power of Attorney or a written Authorization Letter. The team must confirm the exact verification logic with the Treasurer's Office.
- A Tax Clearance issuance requires verification of all prior-year tax payments and arrears. The system currently plans to compute back-year dues based on encoded data. Whether any real-time verification of historical physical receipts is feasible within the project timeline is not yet decided.
- The team and the Treasurer's Office must agree on how long the office will run both the existing manual workflow and the new system at the same time. This comparison period provides data for the Paired t-test evaluation. A duration of two weeks is proposed.
- The team must confirm the exact Assessment Roll format that the Treasurer's Office expects from Module 1. This ensures that the import function successfully processes the incoming data schema without compatibility errors.

7. Attachment of Data Gathering Evidences

Interview Questions
Office Setup & Staff Roles
● How many staff members currently work in the Municipal Treasurer's Office, and what are their main roles (such as collection clerks, cashiers, or billing officers)? ● On average, about how many tax payments or client collections does your office handle in a typical day, and during peak deadline months (such as the first-quarter payment rush)?
Daily Workflows & Tax Computation
● Can you walk us through the step-by-step process from the moment a taxpayer arrives at your window to pay their real property tax until they receive their official receipt or tax clearance? ● How do you compute real property taxes, basic tax (up to 1%), and the Special Education Fund (SEF) levy from the Assessor's Roll? ● What exact discounts are given for prompt/advance payments (e.g., December advance payment), and what interest or penalties are charged for late payments (e.g., monthly delay penalties up to the 72% cap)? ● Where and how are your payment ledgers, tax rolls, and collection records currently stored (physical logbooks, manual ledgers, or a mix of both)? ● How do personnel search for an existing taxpayer record or billing history when a

client inquires about their balance?
Bottlenecks, Delays, & Common Errors
• On average, how long does it take to complete one full payment transaction from initial look-up to receipt printing? • During busy peak seasons, what are the biggest causes of long lines and payment delays at the Treasury counter? • How do you handle taxpayers who arrive with multi-year back-taxes or uncollected arrears? • How often do clerical or calculation mistakes happen during manual tax/penalty computations, and how are they corrected? • Which transactions (like requesting a Tax Clearance for property transfers) require clients to return repeatedly due to outstanding balances or missing payment records? • How is data or coordination handled between the Assessor's Office (when they submit the Assessment Roll) and your office?
Record Keeping & System Integration
• Does your office currently use any software, spreadsheets, or electronic database for tracking tax collections, billings, or delinquencies? • How do you track and flag delinquent properties that have missed payment deadlines?
Taxpayer Experience & Digital Portal Feedback
• What are the most frequent questions or points of confusion property owners express regarding their tax bills or payment deadlines? • Have taxpayers expressed interest in being able to check their billing status or balances online from home before visiting? • What concerns or challenges do you foresee if a web-based billing and collection record system is introduced? • Would your office be open to testing a prototype system alongside your manual books to check its billing speed and computation accuracy?

Sample Documents

REQUEST FORM

Date: _____

WILSON U. LLAMADO
Municipal Assessor
LGU-Pasacao, Camarines Sur

Sir:

May I request from your good office for

- ☐ Certificate of Land holdings
- ☐ Certificate on Non-Improvements
- ☐ Certified True Copy of Tax Declaration
- ☐ Certification of Effectivity
- ☐ Transfer/Revision/cancellation of Real Property/ies

Reason:

_____

_____

Name of Barangay: _____
Tax Declaration No. / PIN: _____

Name of Owner: _____
Address: _____
Contact Number: _____

_____
Requesting Party (Signature over Printed Name)

Approved by:

WILSON U. LLAMADO
Municipal Assessor

MPDO office
ENGR office
APPRAI -

OFFICE OF THE MUNICIPAL ASSESSOR
Canaman Camarines Sur
oooOooo

TRANSACTION SLIP

Control No. : _____
Date : _____
Time : _____

NAME OF REQUESTING PARTY : _____
ADDRESS : _____
NAME OF PROPERTY OWNER : _____
PROPERTY LOCATION : _____
CONTACT NUMBER : _____

A.R.P NO. : _____ PIN NO. : _____
OR NO. : _____ DATE : _____ AMOUNT : _____

- ☐ TRANSFER
- ☐ REVISION
- ☐ SEGREGATION
- ☐ OCULAR INSPI
- ☐ RECLASSIFICA
- ☐ Cancellation of :
- ☐ ANNOTATION
- ☐ NEW DECLARAT

CERTIFICATION
- ☐ LANDHOLDINGS
- ☐ NO IMPROVEMENT
- ☐ WITH IMPROVEMENT
- ☐ HISTORY
- ☐ TAX. DEC
- ☐ CLAV
- ☐ NO PROPERTY

☐ LOT ☐ BUILDING ☐ MACHINERY

SUBMITTED DOCUMENTS
- ☐ E-COPY OF TITLE
- ☐ E-COPY OF REG. DEED OF CONVEYANCE
- ☐ DOAS ☐ EJS W/ SALE ☐ DONATION
- ☐ TRANSFER TAX RECEIPT
- ☐ CERT. AUTHORIZATION REG. (CAR)

- ☐ TAX CLEARANCE (CURRENT YEAR)
- ☐ DAR ORDER/CLEARANCE FOR AGRI. LAND
- ☐ MASTERLIST OF FARMER BENE.
- ☐ APPROVED SURVEY PLAN FROM DENR
- ☐ LETTER REQUEST
- ☐ BUILDING PERMIT
- ☐ BUILDING PLAN
- ☐ CERT. OF OCCUPANCY
- ☐ NOTARIZED SWORN STATEMENT
- ☐ SALES INVOICE (PROOF OF ACQUISATION)
- ☐ ZONING CERT. FROM MPDO
- ☐ APPROVED SUBD. PLAN
- ☐ NOTICE OF DELINQUENCY
- ☐ PROCESSING FEE
- ☐ OTHERS : _____

DETAILS OF ACTION :

RENATO DM. BELEN,REA
MUNICIPAL ASSESSOR

PURPOSE : _____
REMARKS : _____
VERIFIED : _____ POSTED/RECORDED : _____
PROCESSED : _____ TMCR : _____
DOC. FILE NO. : _____
RELEASED BY : _____
DATE SIGNED : _____

FLOW OF TRANSACTION
- ☐ LUZ PANGANIBAN
- ☐ KIM VARGAS
- ☐ ELIZABETH CORDOVA
- ☐ MARK BITARA

Documentation (Pictures)	
CANAMAN	**PASACAO**
	
ONLINE CONSULTATION	
	